



ISO/TC 46/SC 9/Working Group 1
for ISO Project 15706: International Standard Audiovisual Number (ISAN)

Web page: <http://www.nlc-bnc.ca/iso/tc46sc9/isan.htm>

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This is the contribution of Alan SCARBORO to be discussed in the London meeting concerning check digit calculation

Check digit for ISAN

The ISAN group has chosen to use the ISO standard 7064 (Data Processing – Check character systems) as its control character system. The specific method to be used of the eight different proposals made in the ISO document will be the ISO 7064, MOD 11, 10. This method is used when dealing with numeric string information and results in a single digit check sum.

This particular check character system is one of those defined as a Hybrid system. All Hybrid systems use two different moduli in the calculation. One modulus is equal to, and the other is one greater than, the number of characters in the character set of the string to be protected. When referring to the check character system being used by ISAN in a brief mode the ISO designation code to be used is 6.

The method of calculating a check character for the fifteen-character ISAN number is the Hybrid system recursive method. In this method the string is processed character by character for left to right. For checking purposes the string is taken as correct if the string plus its check number is congruent to 1 (mod M) where M is equal to 10 (the number of characters in the character set numeric string).

In our exercise the ISAN number to be checked will be: 654987321654987

M and (M+1) will be the two moduli, where the value of M is equal to the number of characters in the character set of the string being tested. This is always an even number. In our case M is equal to 10 and M+1 is equal to 11.

$|M$ will be equal to the Integer remainder after dividing by M; if this is zero then the value M will be substituted.

$|M+1$ is the integer remainder after dividing by (M+1); the remainder is never zero in this operation.

String values being checked

Product

Product Brought Forward

Adjusted Intermediate sum

Adjusted Product carried forward

Intermediate sum

t e p															
1	10	+	6	=	16		6	x	2	=	12		1		
2	1	+	5	=	6		6	x	2	=	12		1		
3	1	+	4	=	5		5	x	2	=	10		10		
4	10	+	9	=	19		9	x	2	=	18		7		
5	7	+	8	=	15		5	x	2	=	10		10		
6	10	+	7	=	17		7	x	2	=	14		3		
7	3	+	3	=	6		6	x	2	=	12		1		
8	1	+	2	=	3		3	x	2	=	6		6		
9	6	+	1	=	7		7	x	2	=	14		3		
10	3	+	6	=	9		9	x	2	=	18		7		
11	7	+	5	=	12		2	x	2	=	4		4		
12	4	+	4	=	8		8	x	2	=	16		5		
13	5	+	9	=	14		4	x	2	=	8		8		
14	8	+	8	=	16		6	x	2	=	12		1		
15	1	+	7	=	8		8	x	2	=	16		5		
16	5	+	6	=	11		1								

Resulting check digit
required to make the
intermediate sum
congruent to 1